

Seeing what you eat

When should a nutrition app refuse to answer?

COMPSCI 760 · Group 1 · Research Project Proposal

Sushant Krishnan (sthi106) · Keanu De Cleene (kdec819) · Nisarg Patel (npat235)

Shreya Tigga (stig804) · Kevin Zou (rzou895)

MOTIVATION

“620 calories.” Should you believe it?

Photo-based calorie apps are already used for real health decisions: diabetes management, clinical diet tracking, weight programmes.

They answer every photo. They never give a range and they never decline.

A confident wrong number is worse than no number.

What it says: 620 kcal

What it should say:

“480–760, nine times in ten.”

or

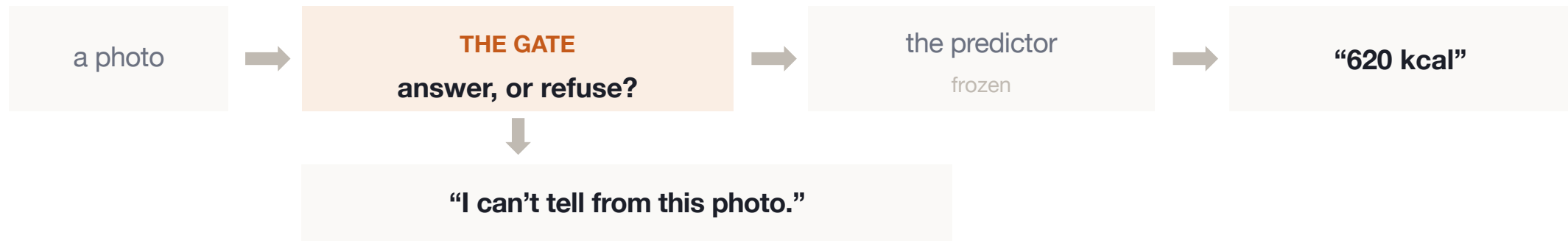
“I can’t tell from this photo.”

To say either of those, an app has to know when it is guessing. We wanted to find out whether it can.

PROBLEM STATEMENT

Everyone is making the number smaller. We want to ask if you can trust it.

We read the food-estimation literature looking for somewhere useful to contribute. Almost all of it reports one number per photo and competes on how small that number is. Very little of it asks whether the number can be trusted. That is the question we picked.



THE PROBLEM TODAY

Deployed models answer every photo with a single number, and are tested mostly on clean lab images. There is no way to decline and no stated confidence.

WHAT WE PROPOSE

Freeze the predictor and study the gate: the part that decides whether to answer. Eight candidate signals, one protocol, with refusing at random as the floor and a perfect refuser as the ceiling.

RESEARCH OBJECTIVES

Three objectives, each with a number attached to it.

Objective	How we measure it	Success criterion	By
O1 · Benchmark Which signals predict when the answer will be wrong?	How well each ranks the errors, on damaged photos and on clean ones	All eight ranked, against random and perfect	11 Sep
O2 · Ceiling How much is left on the table, and on which axis?	Refusing splits two ways: a bad photo, or a hard dish shot well	A gap in kcal for each, with error bars	20 Sep
O3 · Ranges Does the best signal give ranges worth having?	Are the ranges honest for good and bad photos alike; share within ± 100 kcal	Honesty gap ≤ 5 points; usable share reported	11 Oct

All three are read off one results file. That file already exists.

BACKGROUND

Two papers we build on, two protocols we adopt.

Paper	What it did	What it leaves open
[1] Thames et al. Nutrition5k, CVPR 2021	5,006 lab-weighed dishes. End-to-end RGB → calories. 70.6 kcal MAE (RGB), 41.3 kcal (best depth model).	One number per photo, on clean lab images, with no statement of confidence and no way to decline.
[2] Coburn et al. IEEE BHI 2025	Benchmarks 8 large multimodal models on nutrition; metadata and prompting strategies reduce MAE / MAPE.	Reports MAE / MAPE only. No calibration, no ranges, no option to abstain.

WE TAKE ONE IMPROVEMENT FROM EACH

From [1], its evaluation protocol, used as published: we keep the official split, which holds repeated scans of a plate together. A 2026 paper reporting 28 kcal on this dataset re-split it 60/15/25, so we do not think that number is comparable to ours.

From [2], its experimental design, inverted: hold the model fixed and vary the photograph, then report whether the answer can be trusted rather than only how wrong it is.

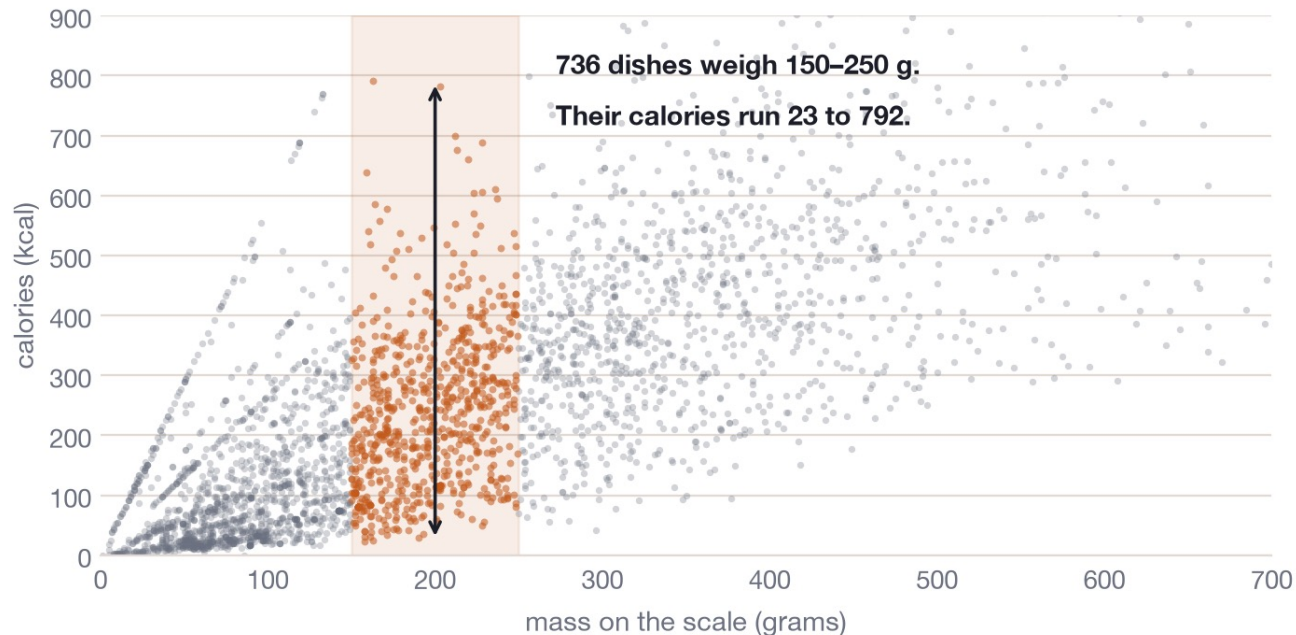
AND TWO PROTOCOLS WE ADOPT RATHER THAN REINVENT refuse the worst x% and measure what is left [3], the biometrics standard since 2007 · damage families × severities [4], the ImageNet-C ladder design

DATASET

Nutrition5k: 5,006 dishes, each weighed on a scale.

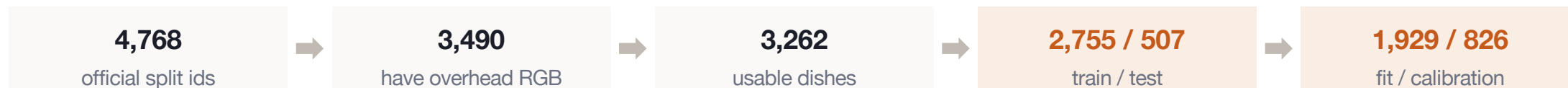


Overhead RGB, one plate, one fixed rig.
Every ingredient weighed beforehand.



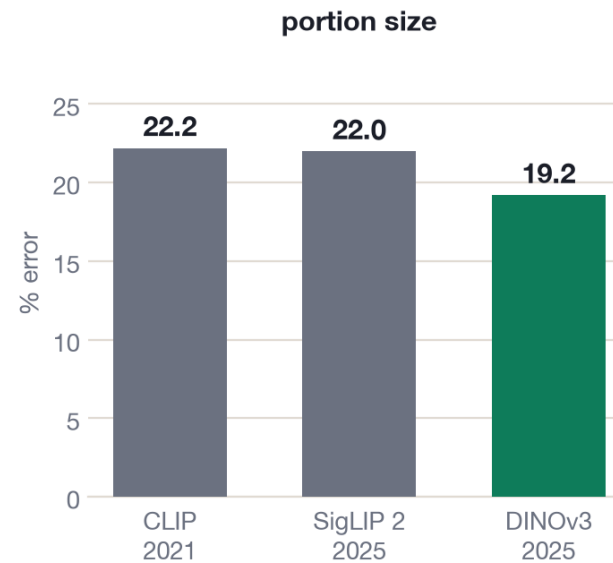
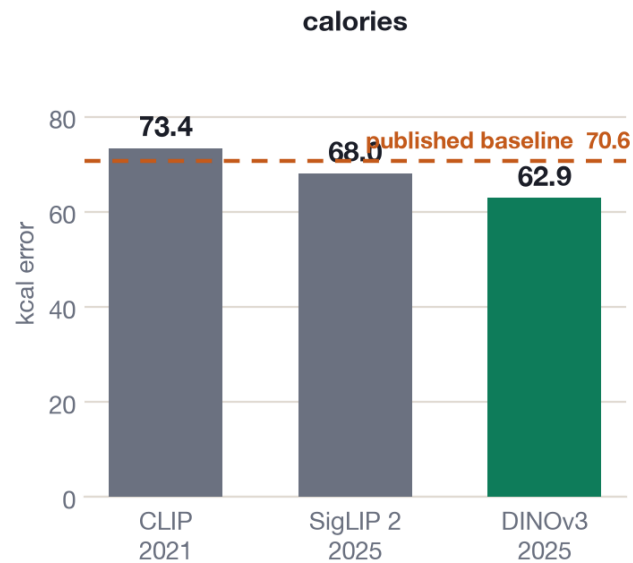
Mass and calories correlate at $r = 0.76$. Weighing the food gets you about half the answer. The rest is knowing what it is.

THE SPLIT WE USE, AND WHAT IT COSTS



The predictor is built. We are not trying to improve it.

THREE FROZEN BACKBONES, RIDGE HEAD, NO TRAINING



507 test dishes, two minutes of compute each. DINOv3 is the only one that improves portion size. We keep CLIP as the primary model because it matches the published baseline, and report the others as a robustness check.

WHY WE ARE NOT CHASING THE NUMBER

A model that named and weighed every ingredient, then made the mistakes we actually measure:

Everything known exactly	33 kcal
Mass off by 10%, names perfect	43 kcal
Mass off by 22%, one name in five wrong	87 kcal

Naming the ingredients separately lands where our one-step model already is. It buys an explanation, not a better number.

The predictor is a fixed control. The open question is whether the app can tell when it is about to be wrong.

Cheap pixel signals do not predict the error.

ONE DISH, THREE PHOTOGRAPHS



clean

sharpness 177

mass error 44 g



blurred

sharpness 1.5

mass error 71 g



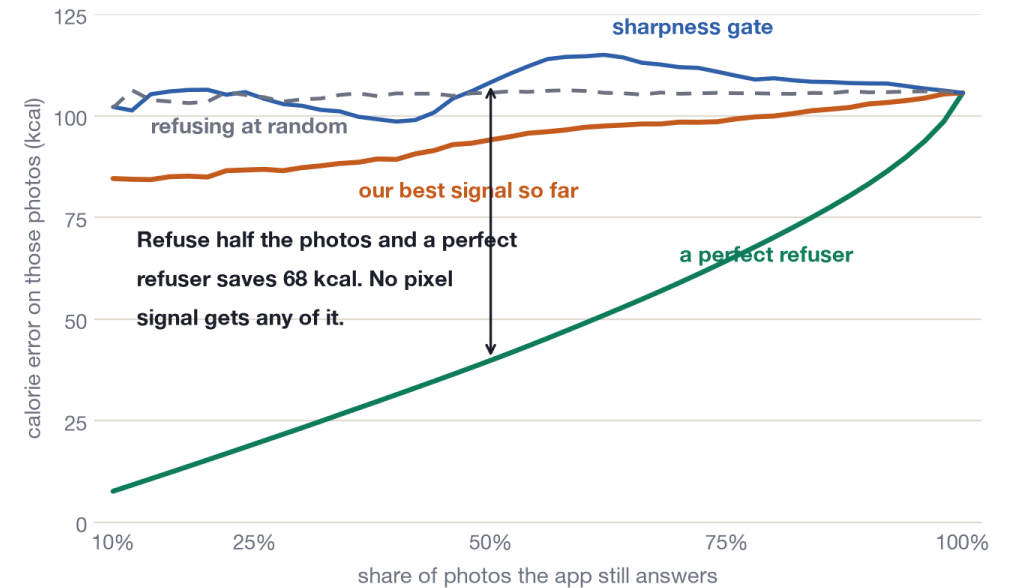
cropped

sharpness 6.5

mass error 167 g

Sharpness calls the cropped photo four times better than the blurred one. It carries two and a half times the error. Cropping removes the plate rim, the only clue to how big the food is.

REFUSE THE WORST PHOTOS. DOES THE ERROR FALL?



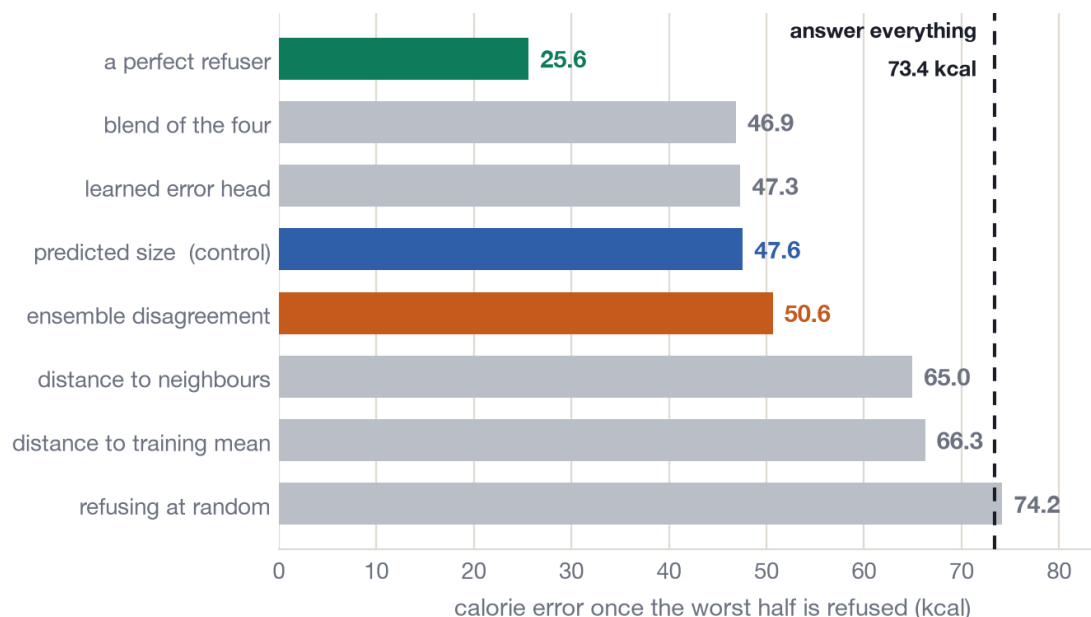
WE SPOT THE DAMAGE ALMOST PERFECTLY sharpness separates blurred, cropped, shrunk and noisy photos from clean ones almost every time ($AUC \approx 1.00$). Seeing damage is not the problem. Damage explains only 7% of the error, and which dish it is explains 56%.

The error is about the dish, not the photograph.

PRELIMINARY RESULT 2 — COMPLETED

Signals that read the model do work. One cheats.

REFUSE THE WORST HALF, CLEAN PHOTOS ONLY



THEN CHECK WHAT EACH ONE REFUSES

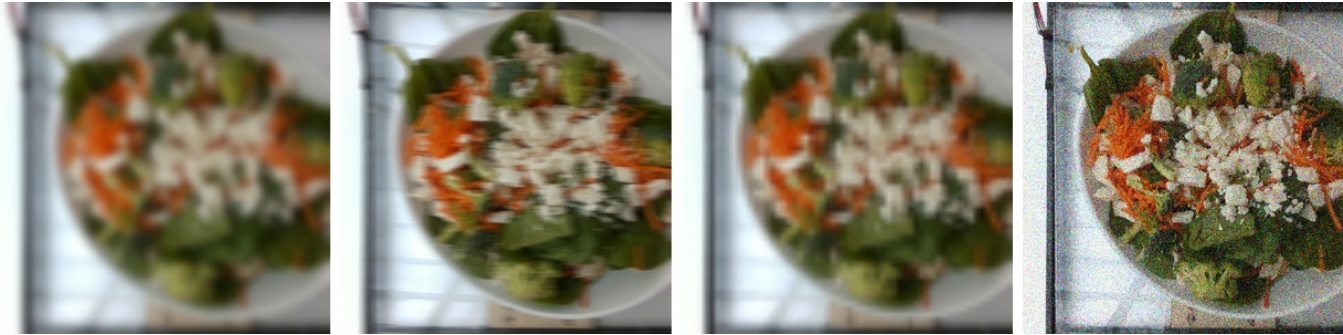


THE CONTROL MATTERS scoring risk by the predicted calorie count alone reaches 47.6 kcal, which looks like one of the best gates. It gets there by refusing 98% of dishes over 400 kcal and answering only for small meals. Error grows with portion size, so any gate can win on error alone by declining big plates.

Ensemble disagreement refuses large meals at the same rate as a perfect refuser, 71% against 72%, and still cuts the error from 73.4 to 50.6.

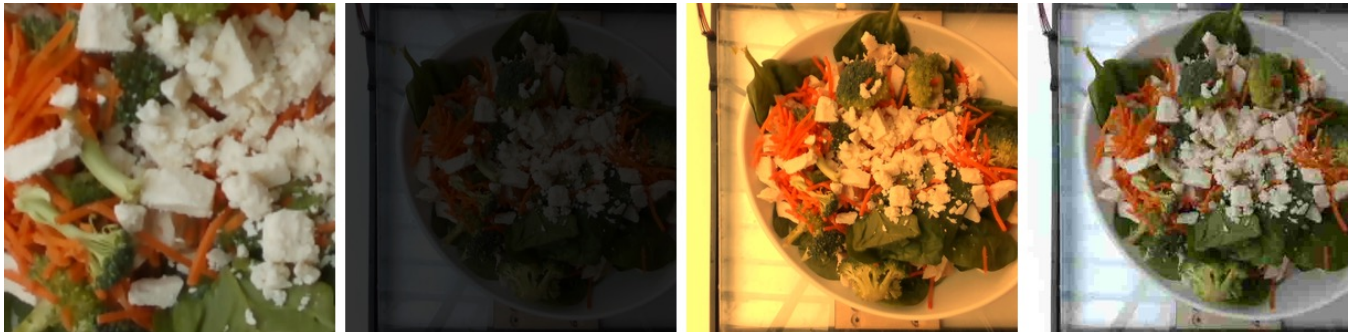
Damage the photos, then score every signal.

DAMAGE A SHARPNESS METRIC CAN SEE



blur · camera shake · shrinking · noise

DAMAGE IT CANNOT



cropping · brightness · colour · compression, plus a phone-camera rung

1 The grouping is the test

The cheap metrics move on the first group and are blind to the second. The error is driven by the second.

2 One results file, written once

Every dish, every damage level, every gate score, in a single CSV. Every question after that is a lookup.

3 Three measurements per gate

How well it ranks the errors, the refuse-and-measure curve [3], and what it does to the width of the confidence range.

Headline question: which signals let an app know it is guessing, and can any of them run on the phone?

TIMELINE & TEAM ROLES

Nine weeks, split so nobody blocks anybody.

When	Milestone	Lead
done — 17 Aug	Data, frozen pipeline, three backbones, first gate results	all
18 – 25 Aug	More damage levels: phone-camera and hidden food	Shreya
26 Aug – 11 Sep	All eight gates scored; the results file	Nisarg
20 Sep	Literature review due (10%)	Sushant
21 – 28 Sep	Update presentation	Keanu
29 Sep – 11 Oct	Confidence ranges: honesty, width, usable share	Kevin
25 Oct	Research report due (22%)	all

WHO OWNS WHAT

Nisarg	Shreya	Keanu	Sushant	Kevin
pipeline + results file	damage levels + pixel gates	model-aware gates + curves	literature + writing	confidence ranges

One results file is the only shared dependency. A member stalling costs one column, not the project.

Every outcome still gives us a result.

Risk

Plan B

No gate beats refusing at random

That is the result, and it is the useful one. It bounds what abstention on a phone can achieve, measured against a ceiling we can compute exactly.

The winning gate turns out to be degenerate

We already found one that is, and caught it with a trivial control. Every gate is reported with what it refuses, not only with its error.

Only the expensive gates work

The cost column is part of the finding. Reliability costing K extra runs is a deployment result, not a failed experiment.

Nutrition5k is one cafeteria, one fixed rig

There is no phone camera in this data, so our phone rung is an imitation and we say so. Claims stay within-distribution, sliced by ingredient rarity rather than cuisine.

The question is which signals work. “None of the cheap ones do” is an answer.

Questions

The error is about the dish, not the photograph.

We can measure that, and we can act on it.

REFERENCES

- [1] Q. Thames, A. Karpur, W. Norris, F. Xia, L. Panait, T. Weyand, and J. Sim, “Nutrition5k: Towards automatic nutritional understanding of generic food,” in Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR), 2021, pp. 8903–8911.
- [2] B. Coburn, J. He, M. E. Rollo, S. S. Dhaliwal, D. A. Kerr, and F. Zhu, “Comprehensive evaluation of large multimodal models for nutrition analysis: A new benchmark enriched with contextual metadata,” in Proc. IEEE Int. Conf. Biomed. Health Informat. (BHI), 2025.
- [3] P. Grother and E. Tabassi, “Performance of biometric quality measures,” IEEE Trans. Pattern Anal. Mach. Intell., vol. 29, no. 4, pp. 531–543, Apr. 2007.
- [4] D. Hendrycks and T. Dietterich, “Benchmarking neural network robustness to common corruptions and perturbations,” in Proc. Int. Conf. Learn. Representations (ICLR), 2019.

All figures, tables and numbers in this deck are our own computation on the Nutrition5k dataset [1] unless cited otherwise.